

1. Administrative & Study Identification

Full Study Title: A TWINSS Extension Trial to Evaluate the Safety and Tolerability of CFZ533 (Iscalimab) at Two Dose Levels Administered Subcutaneously in Patients With Sjögren's Syndrome
Short Title/Acronym: TWINSS Extension **Protocol Number:** CCFZ533B2201E1 **NCT Number:** NCT04541589 **Sponsor Name:** Novartis
Investigational Product: Iscalimab (CFZ533) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of iscalimab at two dose levels (600 mg and 300 mg) administered subcutaneously in patients with Sjögren's Syndrome who participated in the TWINSS core study (CCFZ533B2201).
Secondary Objectives: To further explore the pharmacokinetics (PK) and efficacy of iscalimab at these two dose levels, particularly regarding changes in clinical disease activity (ESSDAI) and patient-reported symptoms (ESSPRI).

2.2 Background & Rationale Sjögren's Syndrome (SjS) is a chronic autoimmune disease characterized by lymphoid infiltration and progressive destruction of exocrine glands, presenting a significant unmet medical need for targeted therapies. This extension study was designed to offer continuation of treatment for participants who successfully completed the TWINSS core study (NCT03905525) and were deemed by the Investigator to clinically benefit from continued iscalimab therapy. The trial provides critical long-term safety, tolerability, and efficacy data for iscalimab (an anti-CD40 monoclonal antibody).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Extension Trial)
Control Type: Dose-comparison (300 mg vs 600 mg regimens)
Blinding: Open-label (participants underwent unblinding after the core study, leading to the discontinuation of placebo)
Randomization: N/A (Participants carry over their respective active allocations from the core study)

3.2 Planned Enrollment & Duration Target \$N\$: Up to 206 participants **Number of Sites:** Global multi-center (rolled over from the core TWINSS trial) **Estimated Study Start:** January 5, 2021 **Estimated Primary Completion:** July 31, 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must have participated in the TWINSS core study, CCFZ533B2201 (NCT03905525), and must have completed the entire treatment period up to Week 48 and the follow-up period up to Week 60. 2. Signed informed consent must be obtained prior to participation in the extension study (i.e., before commencement of the Week 60 assessments of the core study). 3. In the judgment of the Investigator, participants had to be expected to clinically benefit from continued iscalimab therapy.

4.2 Exclusion Criteria 1. Sjögren's Syndrome overlap syndromes where another autoimmune rheumatic disease constituted the principle illness, specifically: Moderate-to-severe active systemic lupus erythematosus (SLE) with anti-dsDNA positivity and renal involvement, active rheumatoid arthritis (RA) impeding ESSDAI articular scoring, or systemic sclerosis. 2. Any other concurrent connective tissue disease (e.g., lupus nephritis, large vessel vasculitis) that was active and required immunosuppressive treatment outside the scope of this trial. 3. Women of child-bearing potential (WOCBP) not using highly effective methods of contraception during dosing and for 14 weeks after stopping the investigational drug.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iscalimab 600 mg or 300 mg administered subcutaneously weekly for the first 3 weeks as loading doses. Subsequently, followed by a bi-weekly (Q2W) maintenance regimen at either 600 mg (2 injections of 300 mg/2 mL) or 300 mg (1 injection of 300 mg/2 mL). **Route of Administration:** Subcutaneous (SC) injection **Duration of Treatment:** 48-week treatment period, followed by a safety follow-up period of 12 weeks (total 60 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care treatments for Sjögren's Syndrome (e.g., symptomatic management of mucosal dryness). **Prohibited:** New concurrent immunosuppressive therapies that would confound the evaluation of Sjögren's Syndrome organ domain assessments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Core Week 60 Informed Consent, Eligibility Confirmation from Core Study, Vital Signs
Baseline	Day 0	Unblinding, First Extension SC Dose, Quality Audit	Interim Weeks 1-48 Bi-weekly SC injections, Safety Labs, PK sampling, Efficacy Assessments (ESSDAI/ESSPRI) End

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of the safety and tolerability of iscalimab, specifically tracked via the incidence of Treatment Emergent Adverse Events (TEAEs). **Measurement Method:** Clinical evaluation and central laboratory analysis up to the last dose date plus 14 weeks (or Week 60).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in the EULAR Sjögren's Syndrome Disease Activity Index (ESSDAI). 2. Change from baseline in the EULAR Sjögren's Syndrome Patient Reported Index (ESSPRI). 3. Pharmacokinetics (PK) profile of iscalimab over the extension period.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring up to 14 weeks post-last dose. An AE is defined as any untoward medical occurrence associated with the use of study intervention. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for severe/life-threatening events. **Data Monitoring Committee (DMC):** Overseen by an internal/external independent review board appointed by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who received at least one dose of iscalimab in the extension phase). **Sample Size Justification:** The sample size is defined by the number of eligible patients successfully completing the CCFZ533B2201 core trial who consent to roll over into the extension (up to 206 participants). **Handling of Missing Data:** Missing ESSDAI or ESSPRI scores from the core study at specified periods strictly observed; standard Missing Data Imputation algorithms as per Novartis SAP protocols.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits to ensure integrity of SC injection compliance and safety recording. **Audit Trail:** Continuous eCRF locking and tracking of unblinded administration via verified data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: CCFZ533B2201E1 **Registry Number:** NCT04541589 **Sponsor/Lead Organization:** Novartis **Study Title:** TWINSS Extension
Official Regulatory Title: A TWINSS Extension Trial to Evaluate the Safety and Tolerability of CFZ533 (Iscalimab) at Two Dose Levels Administered Subcutaneously in Patients With Sjögren's Syndrome

12. Clinical Framework (Sjögren's Specific Adaptation)

Primary Condition: Sjögren's Syndrome (SjS) **Diagnosis Threshold:** Active Sjögren's disease mapped via established core trial parameters (ACR/EULAR criteria). **Medical Methodology:** Biologic Disease Modifying Anti-Rheumatic Drug (anti-CD40 Monoclonal Antibody). **Optimization Target:** Maintenance of ESSDAI reduction and tolerability of long-term therapy.

13. Study Procedures & Methodology

Management Pathway: Subcutaneous self-administration or in-clinic administration of biologic agent. **Education Protocol (HCP):** Unblinding instructions, SC injection training, and adverse event tracking. **Education Protocol (Patient):** Injection site reaction management and disease monitoring. **Quality Audit Mechanism:** Assessment of participant-reported ESSPRI completeness and tracking protocol adherence.

14. Observation & Follow-Up Schedule

Total Duration: 60 Follow-up weeks (48 weeks treatment + 12 weeks safety follow-up) **Onsite Visit Cadence:** Bi-weekly up to Week 48, with a final follow-up at Week 60. **Remote Monitoring Frequency:** Intermittent phone or digital health check-ins as defined by local site capabilities. **Checklist Review Interval:** Routine safety lab reviews at specified intervals aligned with Q2W dosing.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			_____		[DD-MMM-YYYY]								
	Clinical Operations Lead			[Insert Name]			_____		[DD-MMM-YYYY]								
	Lead Statistician			[Insert Name]			_____		[DD-MMM-YYYY]								